Image 1:

```
PS D:\npmtest> node server.js
Node.js web server at port 8000 is running
Nguyen Tan Huy - GCS230465
```

Image 2:

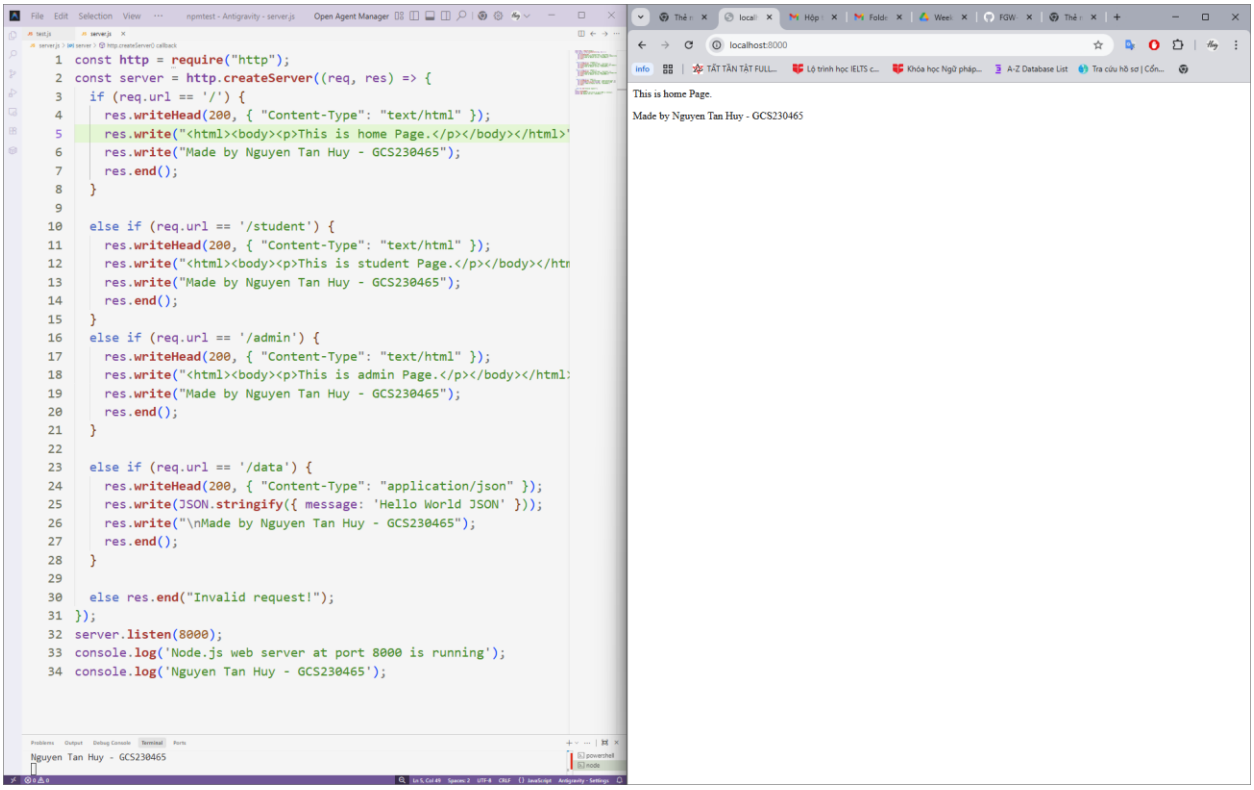


Image 3:

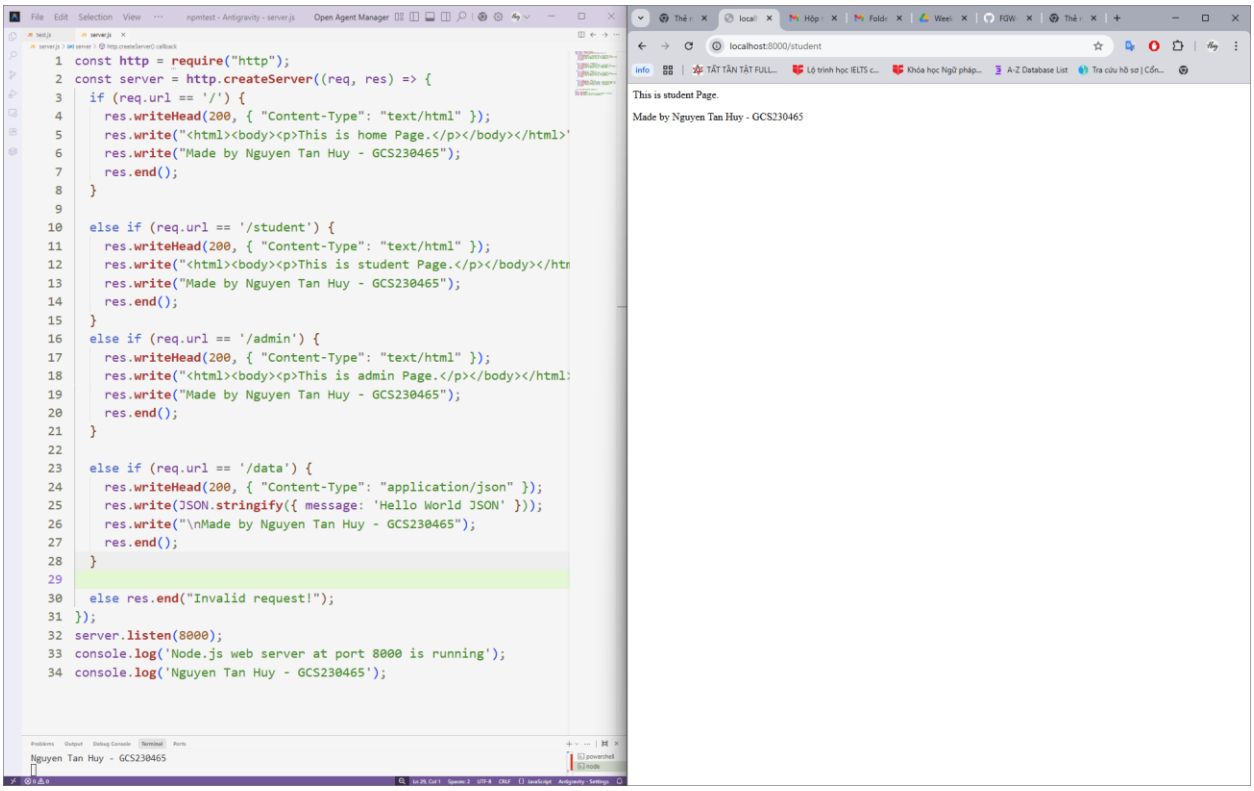
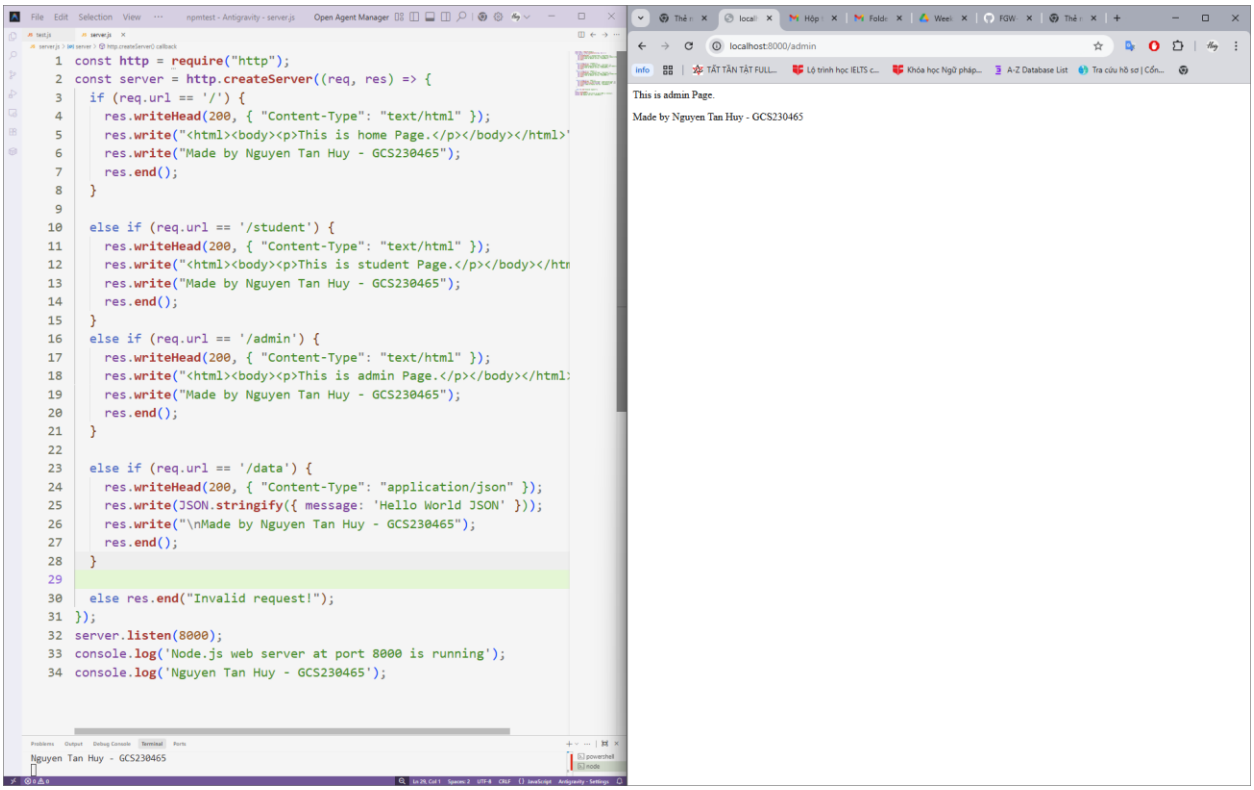


Image 4:



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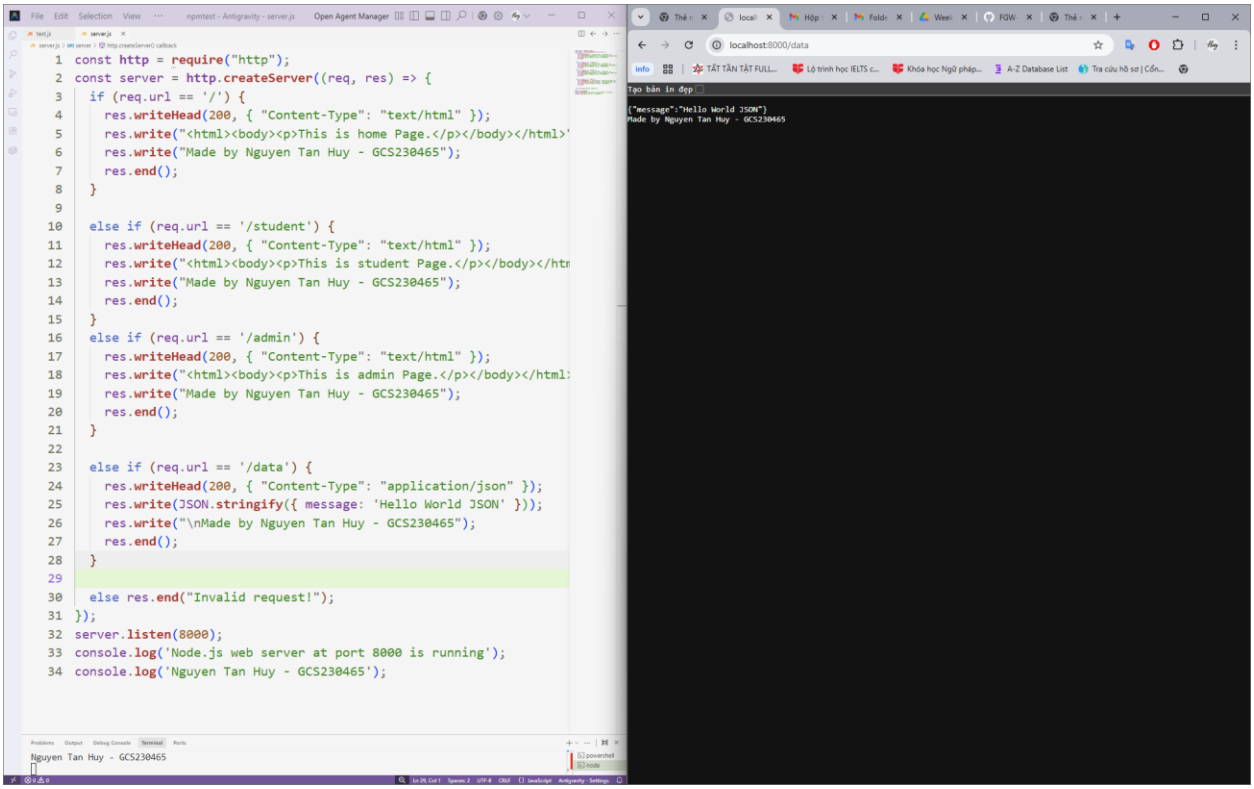
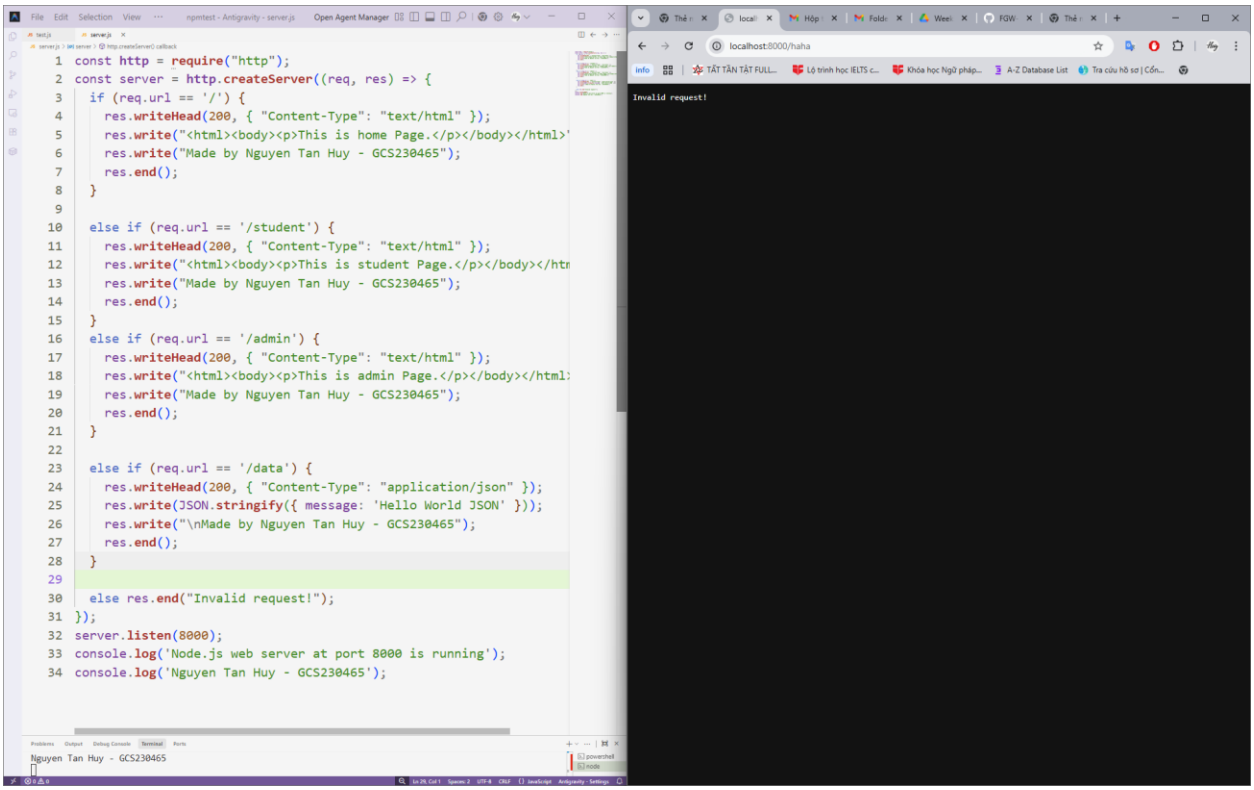


Image 6:



200–300-word explanation of what I have understood about Node.js servers:

My journey into Node.js has been a revelation in how simple yet robust backend development can be. By writing a functional web server from scratch, I have come to appreciate that I don't need heavy frameworks or complex configurations to build something meaningful. With just the built-in http module, I created a system that can listen for incoming traffic on a specific port and decide exactly how to respond based on the user's request.

What makes this approach so powerful is the granular control I have over routing. In my code, I implemented logic to distinguish between a home page, a student page, and an admin page using simple if-else blocks. I also learned how to switch "Content-Types" on the fly—serving structured HTML for users to read, or raw JSON data for other applications to consume. The fact that the server remains responsive and ready for the next request immediately after sending a response (like the "Invalid request!" fallback) demonstrates the efficiency of Node’s non-blocking nature.Beyond the logic, I realized the personal satisfaction of seeing "Node.js web server at port 8000 is running" in my terminal. It proves

that the "barrier to entry" for backend engineering is lower than I thought, while the potential for scaling these small blocks into large, complex systems is nearly limitless.